Friedrich-Alexander-Universität
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Correlation between monthly figures for traffic accidents and weather figures in Münster region.

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Introduction

- Numerous accidents have been reported in recent months
- Adverse weather presents distinct challenges for drivers, affecting visibility and road conditions.
- The country's extensive road network and diverse climate give rise to a intricate interplay between traffic patterns and atmospheric influences.

Is there are relationship between the monthly figures of traffic accidents and the number of weather figures in Munich region?

- Analysing the frequency of accidents involves a comprehensive examination of variables such as month and year, intricately combined with corresponding weather conditions.

- Looking at the development over the time.

Data sources

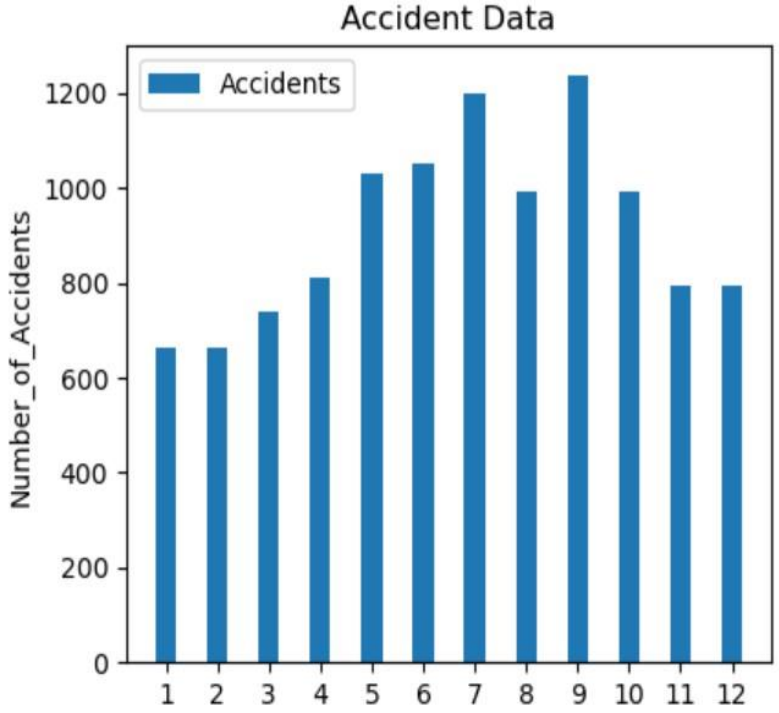
Monthly figures for traffic accidents

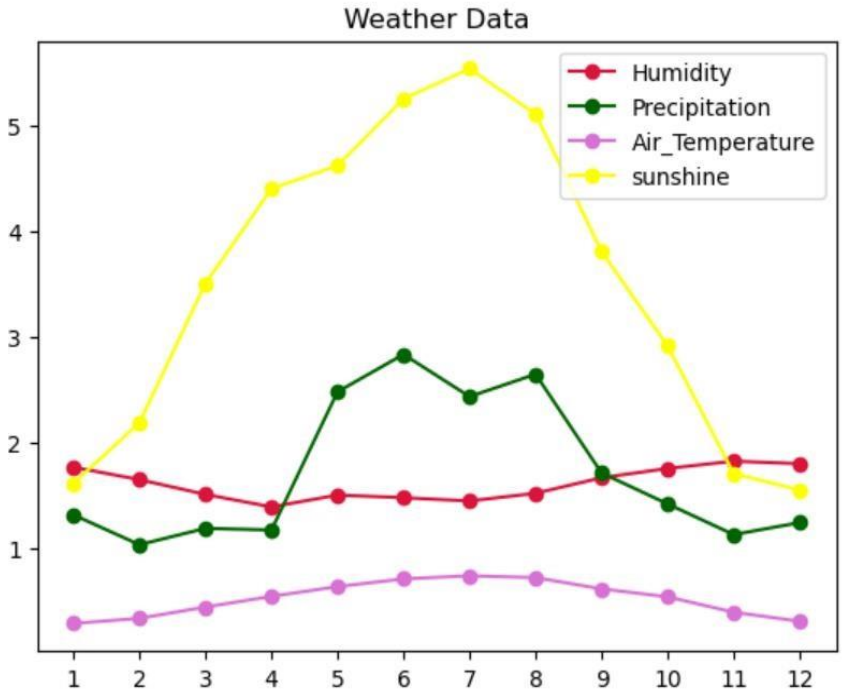
- By Munich Statistical Office
- Provides the number of accidents for Munich region with month and year

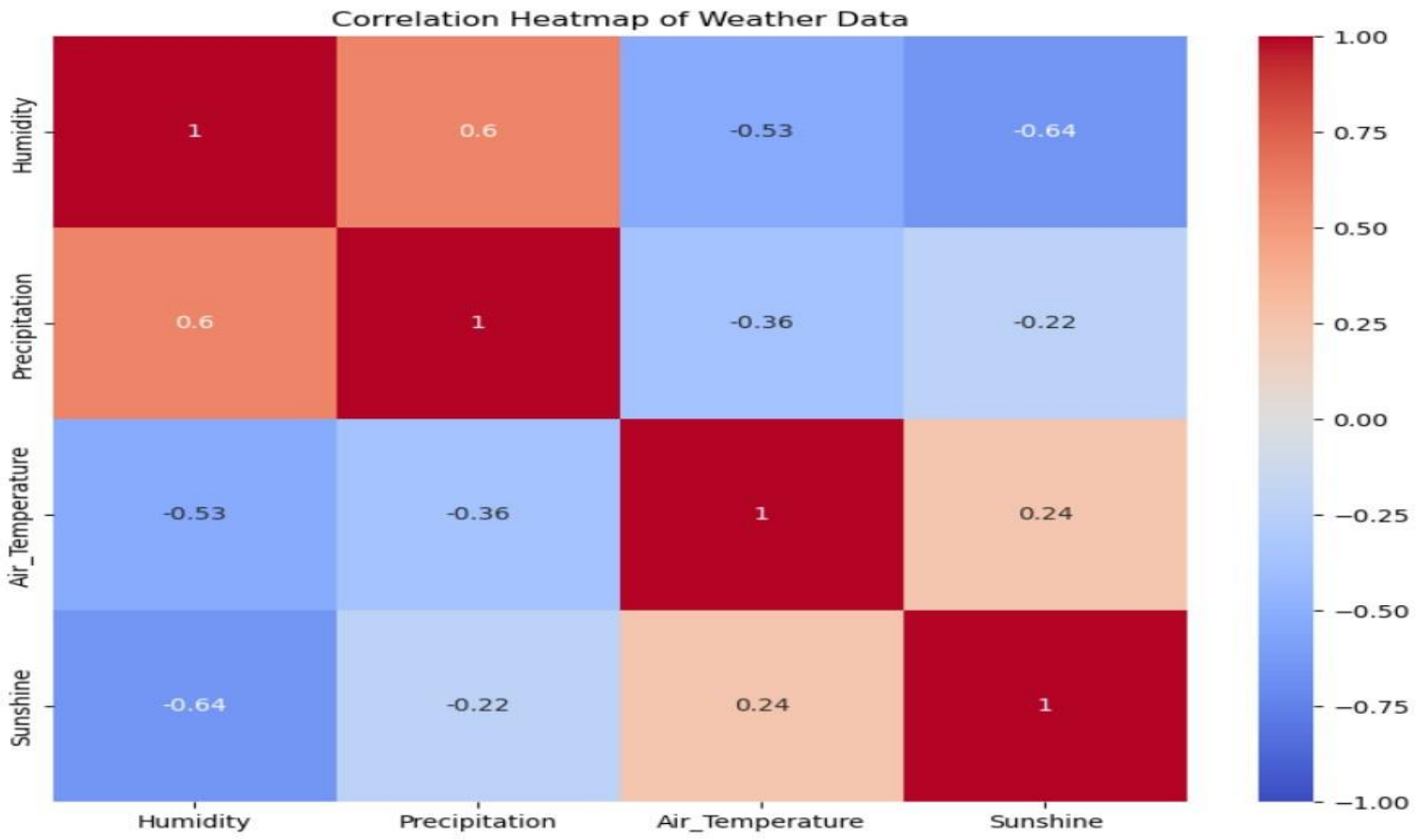
Monthly weather figures

- By Munich Statistical Office
- Provides the monthly weather data with features like humidity, air temperature since year 2000

Results


Accident Data
Accidents
Number_of_Accidents
0
200
400
600
800
1000
1200
1 2 3 4 5 6 7 8 9 10 11 12


Weather Data
Humidity
Precipitation
Air_Temperature
sunshine
1
2
3
4
5
1 2 3 4 5 6 7 8 9 10 11 12

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Pearson Correlation: (-0.3396796871368906, 0.7098640931282802, 0.8525866669633674, 0.7069217592170773)
Spearman Correlation: (-0.44755244755244755, 0.7412587412587414, 0.8531468531468532, 0.7552447552447553)
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Conclusion

- The correlation analysis between Munich's monthly accident figures and weather variables provides crucial insights into their dynamic relationships.
- A good correlation was found between the rate of accident and the weather data, particularly between temperature, humidity, and sunshine exhibit substantial positive correlations, indicating synchronized fluctuations and interconnected dynamics.
- A marked negative relationship is observed between sunshine and precipitation, emphasizing opposing variations in these weather factors.
- Adapting to seasonal variations and weather impacts is vital for ensuring proactive decision-making and maintaining safe transportation in Munich.

Thank you!